SAVEETHA SCHOOL OF ENGINEERING

Assignment – II

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Assignment Topic

A Perceptron-Based System for Detecting Illegal Deforestation Using Satellite Pixel Data (SDG 15 – Life on Land)

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1. Introduction

Illegal deforestation is one of the major causes of biodiversity loss, climate change, and land degradation. Satellite images provide continuous monitoring of forests, making it possible to detect illegal deforestation activities. Machine Learning, particularly the **Perceptron**, can classify satellite pixels as either **Forest** or **Deforested Land** based on features such as the **Vegetation Index (VI)** and **Bare Soil Index (BSI)**.

This application supports **Sustainable Development Goal (SDG) 15 – Life on Land**, which aims to protect, restore, and sustainably manage terrestrial ecosystems.

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2. Problem Statement

A perceptron-based Machine Learning model is designed to classify satellite image pixels into:

- Forest Area
- Illegal Deforested Area

using two input features:

- Vegetation Index (VI)
- Bare Soil Index (BSI)

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3. Objectives

- Understand the perceptron model.
- Explain perceptron representation.
- Explain the threshold (step) activation function.
- Demonstrate how perceptrons classify satellite pixel data.
- Relate the solution to SDG 15.

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4. Perceptron Representation

A **Perceptron** is the simplest type of Artificial Neural Network used for binary classification. It receives multiple input values, multiplies them by weights, adds a bias, and produces an output using an activation function.

Perceptron Model

```
Vegetation Index (x1) ──▶
                          \
                           ▶ Weighted Sum ▶ Threshold Function ▶ Output
                          /
Bare Soil Index (x2) ───▶

Output:
0 = Forest
1 = Deforestation
```

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5. Mathematical Representation

The net input to the perceptron is calculated as:

$$
z = w_1x_1 + w_2x_2 + b
$$

Where:

- (x_1) = Vegetation Index
- (x_2) = Bare Soil Index
- (w_1, w_2) = Weights
- (b) = Bias
- (z) = Net input

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6. Threshold (Step) Function

The perceptron uses a **threshold (step) activation function** to produce the final output.

$$
y =
\begin{cases}
1, & z \geq 0 \\
0, & z < 0
\end{cases}
$$

Where:

- **1** → Illegal Deforestation Detected

- **0** → Forest Area

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7. Example Calculation

Assume:

- Vegetation Index = 0.30
- Bare Soil Index = 0.80

Weights:

- ($w_1 = -0.7$)
- ($w_2 = 0.8$)

Bias:

- ($b = -0.1$)

Step 1: Compute Net Input

$$
z = (-0.7 \times 0.30) + (0.8 \times 0.80) - 0.1
$$

$$
z = -0.21 + 0.64 - 0.10
$$

$$
z = 0.33
$$

Step 2: Apply Threshold Function

Since:

$$
z = 0.33 \geq 0
$$

Output = **1**

Prediction: Illegal Deforestation Detected.

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8. Python Implementation

```
# Perceptron Example

vegetation = 0.30
bare_soil = 0.80

w1 = -0.7
w2 = 0.8
bias = -0.1

net_input = vegetation*w1 + bare_soil*w2 + bias

if net_input >= 0:
    print("Illegal Deforestation Detected")
else:
    print("Forest Area")
```

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9. Advantages of Using Perceptron

- Simple and fast algorithm.
- Efficient for binary classification.
- Easy to implement.
- Suitable for satellite image analysis.
- Supports real-time environmental monitoring.

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10. Limitations

- Works only for linearly separable data.
- Cannot solve complex classification problems alone.
- Sensitive to noisy data.
- Requires appropriate feature selection.

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11. SDG Mapping

SDG 15 – Life on Land

Machine Learning contributes to SDG 15 by:

- Monitoring forests using satellite imagery.
- Detecting illegal deforestation quickly.
- Supporting conservation of biodiversity.
- Helping governments enforce environmental laws.
- Promoting sustainable forest management.

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12. Conclusion

A perceptron-based system provides a simple and effective method for identifying illegal deforestation using satellite pixel data. By analyzing the Vegetation Index and Bare Soil Index, the perceptron classifies land as either forest or deforested area using a weighted sum and a threshold activation function. Although simple, this approach demonstrates how neural networks can assist environmental monitoring and contribute to **SDG 15 – Life on Land** by enabling early detection of illegal deforestation and supporting sustainable ecosystem management.

Viva Questions

1. What is a perceptron?
A perceptron is the simplest artificial neural network used for binary classification.

2. What are the inputs in this system?
Vegetation Index (VI) and Bare Soil Index (BSI).

3. What is the threshold function?
A step activation function that outputs 1 if the net input is greater than or equal to zero; otherwise, it outputs 0.

4. What does the output '1' represent?
Illegal deforestation detected.

5. Which Sustainable Development Goal is related to this application?
SDG 15 – Life on Land.